

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is the server has received too many SYN requests for it to be able to return SYN ACK responses to, so it has crashed.

The logs show that the device with IP address 203.0.113.0 began sending SYN requests at a high rate, to the point where RST ACK responses were given by request number 73, but requests continued to be sent and at a faster rate, causing a server failure.

This event could be a SYN flood attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a SYN request asking to create a session with the server.
2. The server responds with SYN ACK to say it agrees to establish a connection if the response is acknowledged or RST ACK if there is an issue.
3. The client responds with ACK to acknowledge the agreement to the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server must try to respond to all SYN requests with SYN ACK or RST ACK to agree or disagree to establish a connection, but if too many requests are given, the server attempts to respond to them all and fails, leading to a server crash.